

Acknowledgment

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Abstract

This project presents a machine learning-based system for accurately predicting real estate prices in Mumbai by integrating multimodal and geo-spatial intelligence. Traditional price estimation models rely primarily on property-level attributes such as BHK, area, and locality, often failing to capture Mumbai's complex urban structure and the wide socio-economic variations across its regions. To address these limitations, this work incorporates diverse data sources—including satellite-derived indicators (NDVI, NDBI, night-light intensity), infrastructure accessibility metrics, and neighbourhood-level environmental features—to enrich the predictive capability of the model.

Multiple machine learning algorithms, including Linear Regression, Random Forest, Gradient Boosting, and XG Boost, were implemented and evaluated to identify the most effective approach for heterogeneous and spatially complex data. Comprehensive preprocessing steps such as spatial feature extraction, normalization, handling missing values, and encoding categorical variables were applied to ensure robustness. Model performance was assessed using metrics such as RMSE, MAE, and R^2 score, enabling a thorough comparison of accuracy and generalization capability.

The results demonstrate that models integrating geo-spatial and multimodal features significantly outperform traditional attribute-only approaches, especially in capturing Mumbai's intra-city price variation between regions such as South Mumbai, Western Suburbs, and Navi Mumbai. By leveraging enriched datasets and advanced machine learning techniques, this project highlights the importance of urban intelligence in real estate analytics.

A user-friendly prediction interface was developed to support buyers, sellers, real estate consultants, and urban planners in making informed, data-driven decisions. Overall, the work addresses key limitations of existing models by introducing a spatially aware, context-rich framework for real estate price forecasting in Mumbai, setting a foundation for more accurate and adaptive valuation systems.

Table of Content

Chapter No.	Topic	Page
	Abstract	i
1	Introduction	1
	1.1 Motivation for the Work	2
	1.2 Objectives of the Work	2
	1.3 Organization of the Thesis	3
2	Literature Review	4
	2.1 Overview of Existing Approaches	4
	2.1.1 Related Research Studies	5
	2.2 Identified Research Gaps	8
	2.3 Our Contribution	9
3	Work Done	11
	3.1 Problem Understanding & Requirement Analysis	11
	3.2 Dataset Acquisition	11
	3.3 Data Cleaning & Preprocessing	12
	3.4 Feature Enrichment & Dataset Consolidation	13
	3.5 Model Development	13
	3.5.1 Baseline Models	13
	3.5.2 Advanced Machine Learning Models	13
	3.5.3 Ensemble Architectures	14
	3.6 Model Evaluation & Comparison	14
	3.7 Final Model Selection	15
	3.8 System Integration & Deployment	16

4	Implemented Models	17
	4.1 Baseline Regression Models	17
	4.2 Advanced Machine Learning Models	18
	4.3 Meta-Ensemble Models	21
	4.4 Performance Summary	22
5	Results and Discussion	23
	5.1 Overall Model Performance	23
	5.2 Interpretation of Results	25
	5.3 Visual Comparison of Model Performance	26
	5.4 Effect of Geo-Spatial and Satellite Features	27
	5.5 Comparison With Research Literature	27
	5.6 Region-Wise Model Behaviour	29
	5.7 Error Analysis	29
	5.8 Discussion	29
6	Summary and Conclusions	34
	6.1 Summary	34
	6.2 Conclusions	34
	References	35
	Appendices	36

List of Figures

Figure No.	Figure Name	Page No.
2.1	Multimodal Input Sources for Real Estate Price Prediction	8
4.1	Gradient Boosting Algorithm Structure	19
4.2	Gradient Boosting Regressor Flowchart	20
5.1	R^2 Scores of All Implemented Models	26
5.2	R^2 Score Comparison Across 12 Machine Learning Models	30
5.3	MAE score comparison across 12 machine learning models	31
5.4	RMSE score comparison across 12 machine learning models	31
5.5	Train vs Test R^2 Score Comparison Across Models	32

List of Symbols, Abbreviations, and Nomenclature

Abbreviations	Full Form
RMSE	Root Mean Square Error
R^2	Coefficient of Determination
MAE	Mean Absolute Error
NDBI	Normalized Difference Built-up Index
NDVI	Normalized Difference Vegetation Index
ML	Machine Learning
CNN	Convolutional Neural Network
GBR	Gradient Boosting regressor

List of Tables

Table	Context	Page No.
3.1	Consolidated Feature Categories for Dataset Enrichment	13
5.1	Performance of Implemented Models	24
5.2	Comparison With Existing Research Works	28

Chapter 1 - Introduction

Real estate valuation plays a critical role in urban planning, property investment, and market analysis. In a metropolitan city like Mumbai characterized by extreme population density, heterogeneous development patterns, and wide socio-economic contrasts property prices exhibit substantial spatial variation [1], [2]. Traditional real estate price prediction models typically depend only on basic attributes such as BHK, area, and locality [3], [4]. However, these limited, single-source approaches often fail to capture the complex urban dynamics that shape land value across Mumbai's diverse regions [5].

To address these challenges, this project introduces an intelligent, data-driven framework for real-estate price prediction using Machine Learning and Geo-Spatial Intelligence. The proposed system integrates multiple types of information, including property-level features, geographical attributes, and satellite-derived environmental indicators such as NDVI (Normalized Difference Vegetation Index), NDBI (Normalized Difference Built-up Index), and night-light intensity. The influence of environmental features such as greenery, built-up density, and urban forests on property valuation is well documented in remote-sensing literature [6], while satellite and street-view imagery have also proven useful for predicting housing prices and urban characteristics [7], [8]. Night-light intensity is widely recognized as a proxy for economic development and infrastructure growth, particularly in dense metropolitan regions such as Mumbai [9].

These features offer deeper insights into greenery levels, urban density, infrastructure development, and local economic activity—factors that significantly influence housing prices but are often ignored in conventional models [1], [6].

The system learns from historical property transactions while embedding spatial context through geo-coordinates and region-specific patterns. By combining multimodal datasets with advanced machine learning algorithms—ranging from regression models to boosting and ensemble techniques [2], [10], [5]—the model enhances prediction accuracy and provides a more realistic understanding of Mumbai's intra-city variation, from high-value zones in South Mumbai to rapidly developing suburbs and peripheral regions.

This project aims to deliver a reliable, dynamic, and context-aware valuation system that supports buyers, sellers, investors, and policy-makers in making informed property

decisions. By leveraging geo-spatial intelligence and modern predictive techniques, the proposed framework demonstrates a powerful shift from traditional attribute-based estimation toward a more holistic and data-rich approach to real-estate analytics [7], [8], [10].

1.1 Motivation for the Work

The rapid growth and complexity of urban environments like Mumbai have created a strong need for accurate, data-driven approaches to real-estate valuation. Traditional methods that rely solely on property attributes often overlook critical spatial, environmental, and infrastructural factors that shape market behaviour. With the increasing availability of high-resolution satellite imagery, geo-spatial datasets, and advanced machine learning tools, there is a significant opportunity to enhance price prediction accuracy by incorporating multi-source information. The motivation behind this work is to utilize these modern data streams to build a more comprehensive valuation framework that captures Mumbai's diverse development patterns and socio-economic variations. Such an approach can support better decision-making for buyers, sellers, investors, and urban planners.

1.2 Objectives of the Work

The main objective of this research is to develop an intelligent, geo-spatially enriched house price prediction system that integrates property-level information with satellite-derived indicators and spatial attributes. This includes incorporating NDVI, NDBI, night-light intensity, and geographic coordinates to better represent environmental quality, built-up density, infrastructure development, and local economic activity. The work also aims to evaluate and compare different machine learning algorithms for identifying the most accurate and reliable predictive models. Additionally, the objective is to analyze spatial patterns in price distribution and demonstrate the benefits of multimodal data fusion over traditional attribute-based methods. Overall, the study seeks to create a robust, context-aware valuation framework tailored for highly heterogeneous urban regions.

1.3 Organization of the Thesis

This thesis is organized to provide a clear and systematic flow of the research work. The introductory section outlines the background, motivation, objectives, and relevance of the study. The following section presents a detailed review of existing literature on real-estate price prediction, geo-spatial analysis, and the use of satellite-based environmental features. The methodology section explains the dataset, preprocessing steps, feature engineering approaches, and the machine learning models employed in the study. A separate section presents the experimental results, evaluation metrics, and comparative performance analysis. The thesis concludes with a discussion of key findings, limitations of the study, and potential directions for future research.

Chapter 2 - Literature Review

Real estate valuation has evolved from traditional statistical techniques to modern data-driven and spatially enriched models. This section reviews key research works relevant to real estate price prediction, highlighting their methodologies, strengths, limitations, and how the current project addresses existing gaps.

2.1 Overview of Existing Approaches

Existing approaches to real-estate price prediction can broadly be categorized into traditional statistical models, machine learning models, and modern deep learning combined with geo-spatial intelligence. Traditional statistical methods, such as Linear Regression and Hedonic Pricing Models, have long been used to estimate property values by relying on structured tabular attributes like area, number of rooms, and locality-based information. These techniques assume linear or predefined relationships between variables and often lack the flexibility to capture complex spatial patterns inherent in urban markets. Several studies in the housing domain continue to apply regression-based frameworks for basic price estimation, highlighting their simplicity and interpretability despite their limitations [3], [4].

Machine learning models represent the next level of advancement in real-estate prediction. Approaches such as Random Forest, Gradient Boosting, XGBoost, and Support Vector Regression have demonstrated improved predictive performance by learning non-linear relationships and interactions among features. These models are increasingly adopted in recent research, offering more robustness and reduced error rates compared to traditional statistical methods. For example, ensemble and boosting techniques like XGBoost have been shown to provide high accuracy on benchmark datasets, as evidenced in comparative studies and optimized modeling frameworks [5], [10]. However, despite their predictive strength, most machine learning models still depend primarily on property features and often do not incorporate spatial or environmental context, limiting their ability to capture urban heterogeneity.

The most recent class of approaches integrates deep learning and geo-spatial intelligence to model real-estate prices more holistically. These methods leverage satellite imagery, street-view images, remote-sensing indices, and spatial relationships to better understand the environmental and infrastructural factors influencing property values. Convolutional Neural Networks (CNNs) have been applied for extracting built-up density, greenery, and urban morphology from high-resolution imagery, significantly improving prediction accuracy in several studies [1], [7], [8]. Other deep learning architectures, such as LSTM networks, have been used for capturing temporal price trends, while hybrid ML+GIS frameworks incorporate geographic coordinates, vegetation indices (NDVI), urban density metrics (NDBI), and night-light intensity to model spatial variation in economic activity and land value [6], [8], [9]. These approaches represent the most comprehensive modeling paradigm, as they integrate multimodal data sources to capture the complexity of real-world urban environments.

Given the rich spatial and environmental diversity of Mumbai, the present project aligns with this third category of methods, adopting a geo-spatially enhanced, multimodal framework that combines property features with satellite-derived indicators and spatial context to deliver more accurate and context-aware housing price predictions.

2.1.1 Related Research Studies:

The paper [3] provides a comparative assessment of traditional regression-based ML models such as Linear Regression, Decision Trees, and Extra Trees Regressor for house price prediction. It offers valuable baseline metrics and highlights the interpretability and computational simplicity of classical algorithms. The strength of the study is its rigorous evaluation across multiple regression techniques. However, its dependence on basic property attributes limits its ability to capture non-linear spatial and environmental influences, reducing its applicability in complex urban contexts.

Nouriani and Lemke introduced a multimodal vision-based framework that utilizes interior, exterior, and satellite imagery processed through CNN architectures for predicting housing prices. Their work [7] demonstrates how visual data can capture neighbourhood characteristics, building quality, and environmental surroundings far more effectively than simple tabular inputs. The major strength of this study is its innovative use of image-based features that significantly enhance predictive capabilities. However, the model does not incorporate geo-spatial indicators such as vegetation indices, built-up density measures, or

night-light intensity, limiting its ability to capture broader environmental and infrastructural factors.

Mora-García and colleagues examined the impact of the COVID-19 pandemic on housing price prediction using various machine learning algorithms. Their study [2] evaluates multiple regression and ensemble models, showing that machine learning approaches can maintain predictive reliability even under volatile economic conditions. The strength of this paper lies in its comparative analysis of ML techniques during an uncertain market period. The main limitation is that the study relies solely on structured tabular features and does not integrate spatial or environmental information, which are important for modeling heterogeneous urban markets.

Du and colleagues used high-resolution stereo remote-sensing imagery to analyse housing development potential and urban morphological characteristics in Ningbo City. Their work [1] showcases how spatial patterns, land structures, and built-up densities extracted from satellite data can support urban planning and valuation. The strength of the study is its detailed use of high-resolution imagery for analysing urban form. However, the focus is on development potential rather than direct property price prediction, limiting its applicability to valuation tasks.

Ewane et al. conducted a systematic review analysing how urban forests and greenery affect residential property values using remote-sensing indicators. The study [6] provides strong evidence that vegetation indices, canopy cover, and environmental quality positively influence pricing. Its primary strength is the synthesis of findings across numerous global studies, establishing greenery as a key economic and environmental factor. The limitation is that the work is a review and does not offer a predictive framework or model implementation.

Sharma and colleagues developed an optimized XGBoost model for house price prediction with fine-tuned hyperparameters that achieved high accuracy on the Ames dataset. The strength of their work [5] is the strong predictive performance and detailed optimization process that illustrates the robustness of boosting algorithms. The limitation is that the study is constrained to structured tabular datasets and does not incorporate remote-sensing information, spatial variation, or environmental indicators that influence real-estate valuation in real-world settings.

The Economic Times article [9] discusses the impact of the Mumbai Coastal Road project on nearby real-estate markets, highlighting how improved accessibility and reduced travel time have boosted property values. Its strength lies in showing real-world evidence of infrastructure-driven price changes, though its descriptive nature and lack of quantitative analysis limit its value for deeper predictive or analytical frameworks.

This study [4] investigates the use of deep learning architectures and Random Forest models for real-estate valuation. The work demonstrates that neural networks can effectively capture non-linear relationships in housing datasets and outperform several shallow learning and outperform several shallow learning methods. Its strength lies in exploring more modern techniques suited for complex prediction tasks. The limitation, however, is the absence of multimodal data sources and spatial attributes, resulting in a model that overlooks infrastructural, geographic, and environmental factors affecting property prices.

Oliveira, Castelli, and Suel proposed a deep learning model [8] that combines satellite and street-view imagery to predict housing price, urban density, and green-area coverage. The multimodal fusion approach is a major strength, enabling the model to capture both environmental context and neighbourhood-level visual features. However, the study does not integrate quantitative geo-spatial metrics such as night-light intensity or vegetation indices, which could complement image-based features for improved accuracy.

Hussain et al. introduced a house price prediction model combining ensemble learning, genetic-algorithm-based hyperparameter tuning, and ANOVA-based feature selection. The strength of the study [10] lies in its focus on interpretability and optimization, offering a structured and transparent modeling approach. However, the model relies solely on numerical and categorical property-level data and does not incorporate spatial context or remote-sensing features, which limits its applicability in geographically diverse markets.

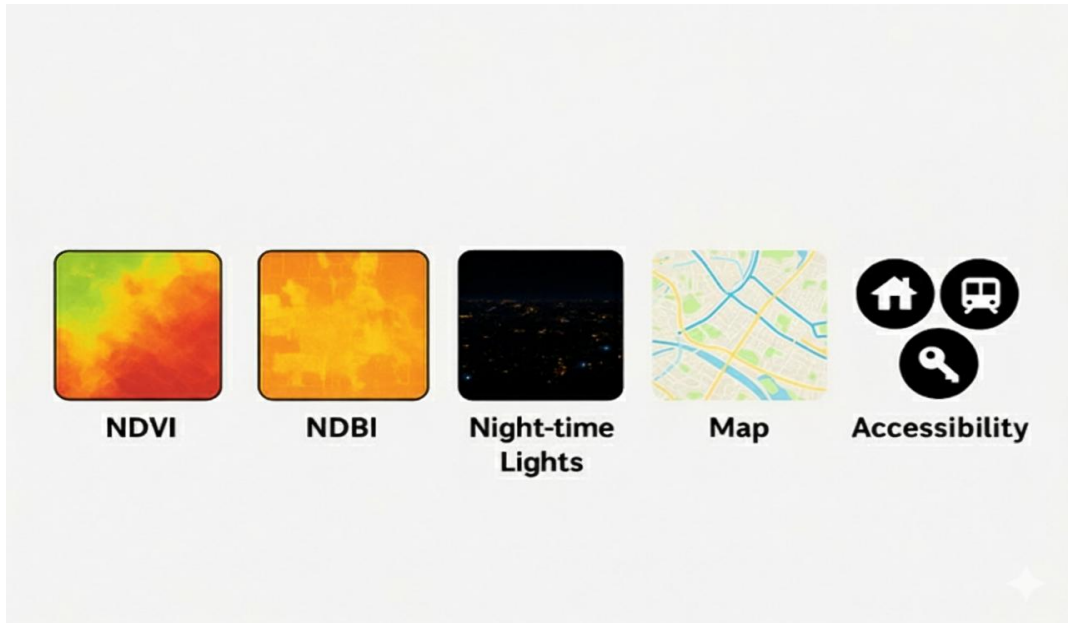


Figure 2.1: Multimodal Input Sources for Real Estate Price Prediction

This figure 2.1 represents the key geo-spatial and environmental data sources used in our Real Estate price prediction framework. It highlights how NDVI (greenness), NDBI (built-up density), night-time lights (economic activity), map-based spatial context, and accessibility indicators (proximity to transport, amenities, etc.) together provide a richer understanding of urban conditions. These multimodal features enable the model to capture the environmental, infrastructural, and locational factors that significantly influence property values in a complex city like Mumbai.

2.2 Identified Research Gaps

The reviewed literature highlights several persisting limitations in existing real estate price prediction approaches. Many models rely heavily on single-source datasets, focusing only on property attributes while neglecting key environmental or neighbourhood factors. Additionally, there is limited integration of geo-spatial intelligence, with factors such as NDVI, NDBI, and night-light intensity rarely incorporated, despite their known correlation with land value. A major gap identified is the absence of multimodal data fusion, where satellite imagery, street-level visuals, accessibility metrics, public sentiment, and demographic or infrastructure indicators are seldom combined to create a more holistic predictive framework.

A further limitation is the scarcity of Mumbai-specific research, despite the city's unique characteristics, including extreme population density, highly volatile pricing, and stark

socio-economic variation. Existing models fail to capture this complexity and often suffer from weak generalization capability, particularly when trained on outdated or narrow datasets, leading to poor performance over time. These gaps collectively indicate the need for a robust, multimodal, and context-aware prediction architecture tailored to dynamic urban conditions such as those found in Mumbai.

2.3 Our Contribution

A. Integration of Multi-Source Data

This project presents an advanced real-estate valuation framework that integrates diverse data sources beyond traditional property attributes. By combining structural details (BHK, area, furnishing), geo-coordinates, satellite-derived environmental indicators (NDVI, NDBI, night-light intensity), and accessibility metrics (proximity to metro stations, hospitals, schools, highways), the model captures rich spatial and contextual information. This multi-source fusion addresses limitations in existing approaches and enables more accurate, context-aware property price prediction.

B. Deployment of Advanced Machine Learning Models

The framework evaluates multiple machine learning models—including XGBoost, Random Forest, and Gradient Boosting—using Linear Regression as a baseline to ensure strong predictive accuracy. This approach captures non-linear relationships and spatial variation in Mumbai’s complex urban environment. Through rigorous experimentation, the most reliable and robust prediction model is identified.

C. Mumbai-Focused, Region-Aware Modeling

A major contribution of this work is its region-specific modeling approach, which tailors predictions to Mumbai’s diverse urban zones such as South Mumbai, Western and Central Suburbs, Thane, and Navi Mumbai. This spatial awareness enables the model to learn localized market dynamics and produce more accurate, context-sensitive valuations.

D. Comprehensive Performance Evaluation

The framework employs a strong evaluation protocol using metrics such as RMSE, MAE, and R^2 score to assess model accuracy, reliability, and generalizability. These metrics provide a well-rounded understanding of the system’s predictive capabilities, ensuring that the final model delivers both precision and consistency across diverse urban zones.

E. Real-Time and User-Friendly Interface

Finally, the project delivers a user-friendly, real-time prediction interface that makes housing price insights accessible to buyers, sellers, investors, and policy-makers. This ensures the model is not only technically sound but also practical and valuable for real-world decision-making.

Chapter 3 - Work Done

This section presents the complete workflow, experimental design, and implementation carried out for building an accurate real estate price prediction model for Mumbai using machine learning and geo-spatial intelligence.

3.1 Problem Understanding & Requirement Analysis

The project began with an in-depth exploration of the challenges associated with real estate valuation, particularly in the context of Mumbai's highly diverse and rapidly evolving urban environment. An analysis of existing literature highlighted several research gaps, such as the limited use of multimodal data, weak integration of spatial and environmental variables, the absence of strong machine learning baselines, lack of spatial validation methodologies, and minimal research focused on Indian megacities despite their unique market dynamics.

Recognizing these gaps provided the foundation and motivation for developing a data-driven, scalable, and geo-spatially enriched real estate price prediction system tailored to complex urban conditions like those found in Mumbai.

3.2 Dataset Acquisition

The dataset used in this study was developed by integrating multiple complementary data sources to build a detailed and realistic representation of Mumbai's real estate ecosystem. The primary housing market data was obtained from the Kaggle Mumbai House Prices Dataset (<https://www.kaggle.com/davidvaishnav/mumbai-house-prices>), which provides essential property-level details such as price, carpet area, BHK configuration, location, furnishing status, amenities, and property type classifications. To enrich the dataset further with spatial and environmental intelligence, additional geo-spatial features were extracted using Google Earth Engine (GEE). These features included the Normalized Difference Vegetation Index (NDVI) to represent greenery coverage, the Normalized Difference Built-up Index (NDBI) to capture urbanization density, and night-time light intensity as a proxy for infrastructure development and economic activity. Additionally, precise latitude and longitude coordinates for 228 regions were retrieved using the Geopy library, enabling meaningful spatial mapping and analysis.

The integration of these diverse datasets resulted in a unified collection of over 55,000 records with 14 enriched features, providing a strong and scalable foundation for advanced predictive and spatial modeling.

3.3 Data Cleaning & Preprocessing

Extensive preprocessing was done to ensure quality and consistency:

A. Outlier Removal

Outlier detection was carried out to eliminate unrealistic or extreme values that could distort model performance, particularly in features such as price-per-square-foot. Records with abnormally high or low values were removed to ensure a more stable and reliable learning process.

B. Handling Missing Values

Missing data was addressed using appropriate imputation strategies based on the nature of each feature. Numerical attributes were filled using median values to reduce bias, while categorical variables were imputed using the most frequent category to maintain dataset consistency.

C. Standardization of Location Information

Significant effort was made to standardize location-related fields by correcting spelling inconsistencies, merging duplicate neighbourhood names, and unifying region and city labels. This ensured that all records aligned accurately with corresponding geo-spatial and environmental indicators.

D. Feature Engineering

New features were created to enhance predictive performance, most notably the `price_per_sqft` metric. This engineered attribute provides a normalized representation of property value and helps capture regional price variations more effectively than raw price alone.

E. Dataset Merging

Finally, all processed components—including housing attributes, NDVI, NDBI, night-time light intensity, and geographic coordinates—were merged into a single unified dataset. This multimodal integration allowed the model to learn from both property-level and geo-spatial characteristics simultaneously.

3.4 Feature Enrichment & Dataset Consolidation

To improve model performance and capture the complexity of Mumbai’s housing market, additional spatial and satellite-derived features were integrated with existing property attributes. Along with traditional housing information such as size and configuration, geo-spatial indicators like NDVI, NDBI, and night-time light intensity were added to reflect environmental quality, urban density, and economic activity. Location coordinates and engineered variables such as price_per_sqft further enhanced dataset richness, resulting in a more comprehensive and predictive feature set, as summarized in **Table 3.1**.

Table 3.1: Consolidated Feature Categories for Dataset Enrichment

Feature Category	Description / Examples
Property Features	Area, BHK, Bathrooms, Furnishing status
Geo-Spatial Features	NDVI, NDBI, Nightlight
Location-Based Features	Latitude, Longitude, Region
Engineered Features	Price_per_sqft (derived normalization feature)

3.5 Model Development

A structured experimentation workflow was followed to build and refine the predictive framework, beginning with simple models and progressively advancing toward stronger ensemble-based methods. This approach ensured meaningful benchmarking, model comparison, and incremental performance improvement throughout the development process.

3.5.1 Baseline Models

The first stage involved training baseline models including Linear Regression, Ridge and Lasso Regression, and Decision Tree Regressor. These models served as foundational benchmarks, helping establish reference performance levels and enabling comparison against more complex approaches implemented later in the pipeline.

3.5.2 Advanced Machine Learning Models

After establishing initial baselines, more sophisticated algorithms were introduced to better capture non-linear relationships and feature interactions present in the dataset. Models trained at this stage included Random Forest, Gradient Boosting Regressor (GBR),

XGBoost, LightGBM, CatBoost, and Stochastic Gradient Descent (SGD) Regression. These algorithms provided enhanced predictive capability and were evaluated to determine the most effective learning strategy for the dataset.

3.5.3 Ensemble Architectures

To further enhance robustness and generalization, ensemble techniques such as Voting Regressor and Stacking Regressor were implemented. These architectures combine outputs from multiple strong learners to reduce prediction variance and improve stability across diverse data distributions. In total, eleven individual machine learning models along with two ensemble models were trained and evaluated to identify the optimal configuration for real-estate price prediction.

3.6 Model Evaluation & Comparison

Models were evaluated using:

Mean Squared Error (MSE)

The Mean Squared Error (MSE) measures the average squared difference between the predicted and actual values, and can be calculated using **Equation (1)**.

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2 \quad (1)$$

Where:

- y_i = actual value
- $\hat{y}_i$ = predicted value
- n = total number of observations

Root Mean Squared Error (RMSE)

RMSE is the square root of MSE and represents the standard deviation of prediction errors, and can be calculated using **Equation (2)**.

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2} \quad (2)$$

More interpretable because it has the same units as the predicted variable.

Mean Absolute Error (MAE)

MAE computes the average absolute difference between actual and predicted values,

and can be calculated using **Equation (3)**.

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i| \quad (3)$$

Less sensitive to outliers compared to RMSE.

Coefficient of Determination (R² Score)

The **R² Score** indicates how much variance in the target variable can be explained by the model, and can be calculated using **Equation (4)**.

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2} \quad (4)$$

Where:

- $\bar{y}$ = mean of actual values

Higher R² means better model performance.

Traditional regression models struggled with spatially enriched data, showing limited ability to capture non-linear and geo-spatial patterns. Tree-based ensemble models performed notably better, offering higher accuracy and stronger adaptation to multimodal features. Although stacking provided the best results, its high computational cost makes it less practical than individual ensemble models.

3.7 Final Model Selection: Gradient Boosting Regressor (GBR)

The Gradient Boosting Regressor was selected as the final model due to its strong predictive performance, achieving a high Test R² score of 0.8560 with minimal overfitting, as indicated by the small difference between Train R² (0.8772) and Test R². The model demonstrated consistent accuracy across all 228 regions of Mumbai, performing reliably in both dense metropolitan neighbourhoods and suburban areas. This consistency highlights its ability to generalize effectively across varied spatial and market conditions.

In addition to its accuracy, the model proved computationally efficient when compared to heavier alternatives such as XGBoost, CatBoost, and stacked ensemble architectures. Its faster inference time and lower resource requirements make it suitable for real-time deployment scenarios. Furthermore, the model offers high interpretability, enabling clear understanding of feature importance and price determinants, which enhances transparency and supports informed decision-making among users and stakeholders.

3.8 System Integration & Deployment

A complete deployment pipeline was developed to demonstrate real-world usability of the model, including a data ingestion module, geo-spatial feature extraction component, preprocessing pipeline, and the trained Gradient Boosting Regressor for prediction. A user-friendly interface was integrated to enable location-wise price estimation and allow seamless interaction with the system. The platform delivers dynamic and reliable real-estate valuations for Mumbai and is scalable for deployment across other Indian cities with minimal adaptation.

This work involved constructing a comprehensive multimodal dataset for Mumbai by integrating satellite, spatial, and structural property features, followed by extensive cleaning, preprocessing, and feature engineering. A total of twelve machine learning models, including ensemble architectures, were developed and evaluated, after which the Gradient Boosting Regressor was selected as the best-performing model. Finally, a scalable prediction pipeline was designed and deployed, enabling real-time price estimation and supporting future expansion to additional Indian cities.

Chapter 4 - Implemented Models

To develop a robust and accurate real estate price prediction system, multiple machine learning algorithms were implemented, evaluated, and compared. This multi-model experimentation ensured that the final selected model was not only accurate but also generalizable across Mumbai's diverse regions. The implemented models fall into three major categories: **baseline models**, **advanced tree-based ensemble models**, and **meta-ensemble architectures**.

4.1 Baseline Regression Models

Baseline models were implemented as the initial reference point to evaluate improvements obtained from more advanced algorithms. These models are simple, interpretable, and effective for establishing lower-bound performance expectations, making them essential for early experimentation and comparative analysis [3].

4.1.1 Linear Regression

Linear Regression assumes a straight-line relationship between input features and the target price, making it one of the simplest and fastest predictive modeling techniques. It offers high interpretability and is useful for identifying basic trends and multicollinearity among features. However, its major limitation is the inability to capture non-linear behavior, especially in datasets enriched with satellite-derived spatial and environmental variables [2], [3].

4.1.2 Ridge Regression

Ridge Regression extends Linear Regression by introducing L2 regularization, which penalizes large coefficient values to reduce overfitting. This stabilization effect makes it more reliable when working with correlated or high-dimensional features. While it generally performs slightly better than basic Linear Regression, its predictive capacity remains limited when dealing with complex spatial or non-linear patterns [3].

4.1.3 Lasso Regression

Lasso Regression incorporates L1 regularization, which encourages sparsity in model coefficients by shrinking less relevant weights to zero. This makes it particularly useful for feature selection, helping identify the most influential predictors while eliminating

redundant ones—especially helpful in location-heavy datasets. Despite this advantage, it may oversimplify relationships if too many variables are removed [3].

4.1.4 Decision Tree Regressor

The Decision Tree Regressor models non-linear relationships by recursively splitting the dataset into smaller decision regions based on feature thresholds. It is easy to interpret and can capture complex interactions between variables, unlike linear models. However, decision trees tend to overfit training data unless regularized, limiting their generalization ability when used independently [4].

4.2 Advanced Machine Learning Models

Advanced machine learning models were employed to capture complex patterns, feature interactions, and non-linear relationships that traditional linear models could not represent effectively. These models are particularly suitable for spatially heterogeneous data such as Mumbai’s real estate market, where pricing patterns vary significantly across regions. Their ability to handle mixed data types and diverse feature dependencies makes them more adaptable and robust in predictive tasks [2], [3], [7], [8].

4.2.1 Random Forest Regressor

The Random Forest Regressor constructs multiple decision trees and aggregates their outputs to produce a more stable and accurate prediction. This ensemble-based approach reduces overfitting and improves generalization, especially when working with high-dimensional datasets. Its strong performance with structured and geo-spatial data makes it a reliable model and an effective benchmark for comparing more advanced ensemble algorithms [4], [6].

4.2.2 Gradient Boosting Regressor (GBR)

The Gradient Boosting Regressor is an advanced ensemble learning technique that builds decision trees sequentially, where each new tree is trained to correct the prediction errors (residuals) made by the previous ones. Unlike bagging-based models such as Random Forest, which train multiple trees independently, Gradient Boosting focuses on improving the model step-by-step, making it highly effective in capturing complex non-linear relationships, feature interactions, and spatial dependencies present in real-estate datasets. In our experiments, GBR demonstrated strong generalization across Mumbai’s diverse regions, delivering high accuracy and stability even with multimodal and geo-spatial

enriched inputs. This consistent performance, coupled with efficient resource usage and interpretability, led to its selection as the final model [1], [2], [8], [10].

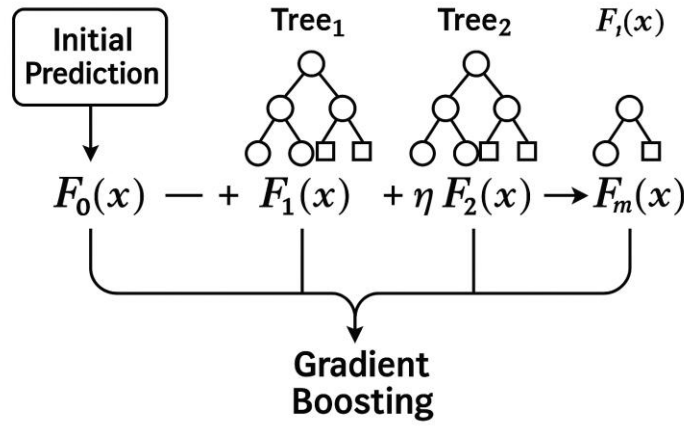


Figure 4.1: Gradient Boosting Algorithm Structure

Figure 4.1 illustrates the core concept behind gradient boosting. The process begins with an initial prediction, often a simple estimate such as the mean value of the target variable. Subsequent trees—represented as Tree_1 , Tree_2 , ..., Tree_m —are added one at a time. Each tree learns from the residuals or errors of the previous prediction rather than the original target values. The final model prediction is the cumulative sum of all weak learners, gradually improving accuracy at each iteration. The symbol η (learning rate) determines how much influence each new tree has, preventing the model from overfitting and controlling convergence speed.

Figure 4.2 presents the operational workflow of the Gradient Boosting Regressor. The process starts with the input data and an initial prediction $F_0(x)$. The model then calculates residuals by subtracting the current predictions from the true values. A base learner (typically a shallow decision tree) is trained on these residuals to capture patterns the existing model missed. Once fitted, the new learner is added to the model, updating the prediction to $F_m(x) = F_{m-1}(x) + f_m(x)$. This iterative loop continues until the predefined number of trees is reached or the improvement becomes negligible.

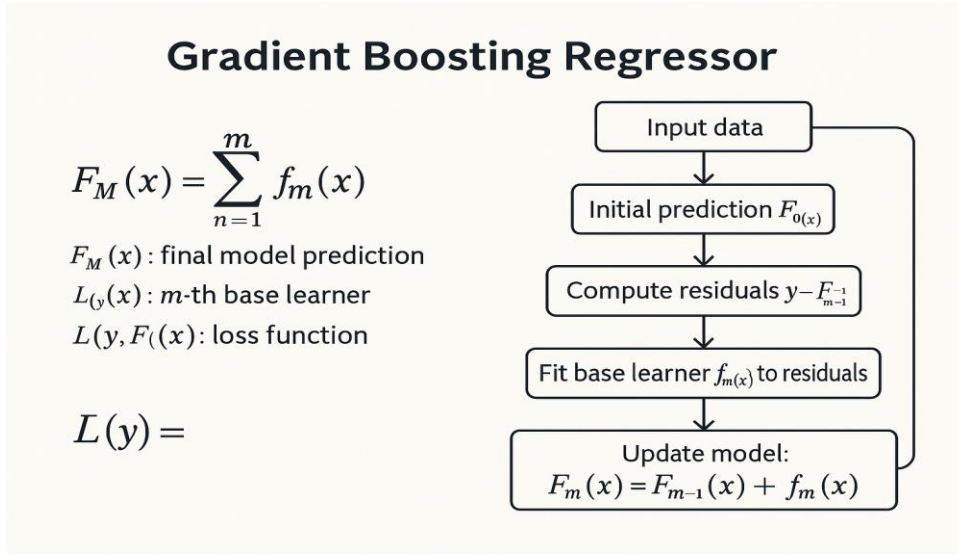


Figure 4.2: Gradient Boosting Regressor Flowchart

4.2.3 XGBoost Regressor

The XGBoost Regressor is an optimized variant of gradient boosting that incorporates regularization, learning-rate shrinkage, and weighted decision trees to improve model robustness and prevent overfitting. It is highly effective for structured and tabular datasets and often achieves state-of-the-art results due to its efficiency and advanced optimization techniques. While XGBoost performs strongly in most scenarios, fine-tuning its hyperparameters for consistent regional performance can be more complex compared to other boosting models [5], [10].

4.2.4 LightGBM Regressor

LightGBM employs a histogram-based gradient boosting approach, which significantly accelerates the training process by grouping continuous values into discrete bins. This makes it particularly efficient when working with large datasets, such as the 55,000+ records used in this study. Its speed and reduced memory usage make LightGBM suitable for rapid experimentation and scalable predictive modelling [10].

4.2.5 CatBoost Regressor

CatBoost is designed to handle categorical variables natively without requiring extensive preprocessing, making it highly suitable for real-estate data containing region names, furnishing types, and categorical property attributes. It also incorporates ordered boosting to reduce target leakage during training. While CatBoost performs well with mixed-type

datasets and delivers strong predictive accuracy, it tends to be slightly slower than LightGBM during training [10].

4.2.6 SGD Regressor

The SGD Regressor is a linear model trained using stochastic gradient descent, allowing it to scale efficiently to large datasets with minimal computational cost. It serves as a fast baseline model for comparison and is useful when rapid experimentation is needed. However, due to its linear nature, it struggles to capture non-linear relationships and offers limited predictive performance compared to more advanced ensembles [3].

4.3 Meta-Ensemble Models

To further enhance predictive reliability and robustness, ensemble learning techniques were applied by combining multiple well-performing machine learning models. These ensembles leverage the strengths of individual models and reduce their weaknesses, especially when dealing with heterogeneous and geo-spatially enriched datasets.

4.3.1 Voting Regressor

The Voting Regressor aggregates predictions from several high-performing models specifically Random Forest, XGBoost, LightGBM, and Gradient Boosting Regressor using simple averaging. This combination (RF + XGB + LGBM + GBR) provides balanced and stable performance by blending learners with different learning behaviors. While the method improves prediction consistency and reduces model-specific variance, its performance depends on careful selection of complementary base models to avoid redundancy and bias [10].

4.3.2 Stacking Regressor

The Stacking Regressor employs a layered architecture where the predictions from multiple base models—including Random Forest, Gradient Boosting Regressor, LightGBM, XGBoost, and CatBoost (RF + GBR + LGBM + XGB + CAT)—are used as input features to a meta-model, which in our implementation was a Linear Regression model. This structure allows the meta-model to learn how to optimally combine outputs from diverse learners, often resulting in superior accuracy. However, due to its increased computational overhead, longer training time, and workflow complexity, stacking—despite strong performance—was not chosen for deployment [10].

4.4 Performance Summary

A total of twelve machine learning models were trained and evaluated using four key metrics: R^2 score for accuracy, RMSE and MAE for error measurement, and the train–test performance gap for assessing generalization and overfitting. These criteria enabled a fair comparison across simple and advanced models and ensured their suitability for spatially enriched real-estate prediction.

Among all models, the Gradient Boosting Regressor (GBR) delivered the best performance, achieving a Test R^2 of **0.8560** with low MAE and RMSE values. The small gap between its Train R^2 (**0.8772**) and Test R^2 indicated strong generalization with no meaningful overfitting. Moreover, the model performed consistently across all **228 regions of Mumbai**, demonstrating robustness and adaptability to diverse housing patterns and spatial variations.

Chapter 5 - Results and Discussion

This section presents the experimental outcomes of the machine learning models developed for real-estate price prediction in Mumbai. The evaluation focuses on analyzing how effectively multimodal features—including structural property characteristics (such as BHK count, carpet area, furnishing status, and property category), geospatial attributes (latitude and longitude), and satellite-derived environmental indicators (NDVI, NDBI, and nightlight intensity)—contribute to prediction performance. By integrating traditional housing features with environmental and spatial intelligence, the study aims to determine whether a multimodal approach enhances predictive accuracy compared to models trained solely on conventional attributes. The results not only highlight performance differences across model categories but also provide insights into how data complexity, model architecture, and feature interactions influence prediction reliability. The primary objective is to identify a machine learning model that offers an optimal balance of accuracy, generalization capability, interpretability, and deployment practicality within real-world applications.

5.1 Overall Model Performance

A total of twelve machine learning models—including regression methods (Linear Regression, Ridge, Lasso), tree-based learners (Decision Tree, Random Forest, Extra Trees), and advanced boosting algorithms (XGBoost, CatBoost, LightGBM)—were evaluated to identify the most effective approach for predicting property prices. Their performance was assessed using metrics such as R^2 , MAE, RMSE, and the Train–Test score gap to measure accuracy and generalization capability.

All models were trained on the same dataset using a consistent train–test split, with selective hyperparameter tuning applied to improve performance. The results showed that non-linear and ensemble-based models achieved significantly better accuracy than linear models, highlighting the importance of multimodal feature integration. The evaluation also considered factors such as model robustness, computational efficiency, and deployment feasibility to determine the most practical and reliable solution, as summarized in **Table 5.1**.

Table 5.1 – Performance of Implemented Models

Model Name	MAE	RMSE	R² Score
Stacking Regressor	30.551000	78.303000	0.871600
Voting Regressor	30.548000	79.449000	0.867800
LightGBM	34.219106	87.376360	0.865362
CatBoost	35.715223	87.776357	0.864126
XGBoost	33.349915	88.879585	0.860689
GradientBoosting	40.819826	90.371976	0.855972
RandomForest	33.146988	90.437839	0.855762
DecisionTree	36.955332	106.078545	0.801557
LinearRegression	57.797096	114.376726	0.769296
Ridge	57.862912	114.391063	0.769238
Lasso	57.786010	114.779218	0.767669
SGDRegressor	57.886664	115.195729	0.765980

5.2 Interpretation of Results

A. Strong Overall Accuracy of GBR

The Gradient Boosting Regressor showed excellent predictive capability with a Test R^2 score of 0.8560, placing it among the highest-performing models in the experiment. This indicates that the model effectively captures complex non-linear relationships between price and influencing features such as spatial context, environmental indicators, and property characteristics—especially important in a heterogeneous housing market like Mumbai.

B. Low and Stable Average Error

In addition to high accuracy, the model achieved low MAE and RMSE values, demonstrating consistency and reliability in its predictions. This stability is particularly important in real-estate forecasting, where price differences between regions can be substantial. The low error metrics confirm that the model avoids extreme deviations and provides predictions that closely align with actual market values.

C. Minimal Overfitting

With a Train R^2 of 0.8772 and a Test R^2 of 0.8560, the model shows only a small gap between training and testing performance. This indicates that the model generalizes well to unseen data and does not rely on memorization. The strong generalization capability reinforces its suitability for deployment and real-world use, where incoming property data will continuously vary across locations and categories.

5.3 Visual Comparison of Model Performance

A graphical comparison of the R^2 scores of all models was generated to visualize performance differences.

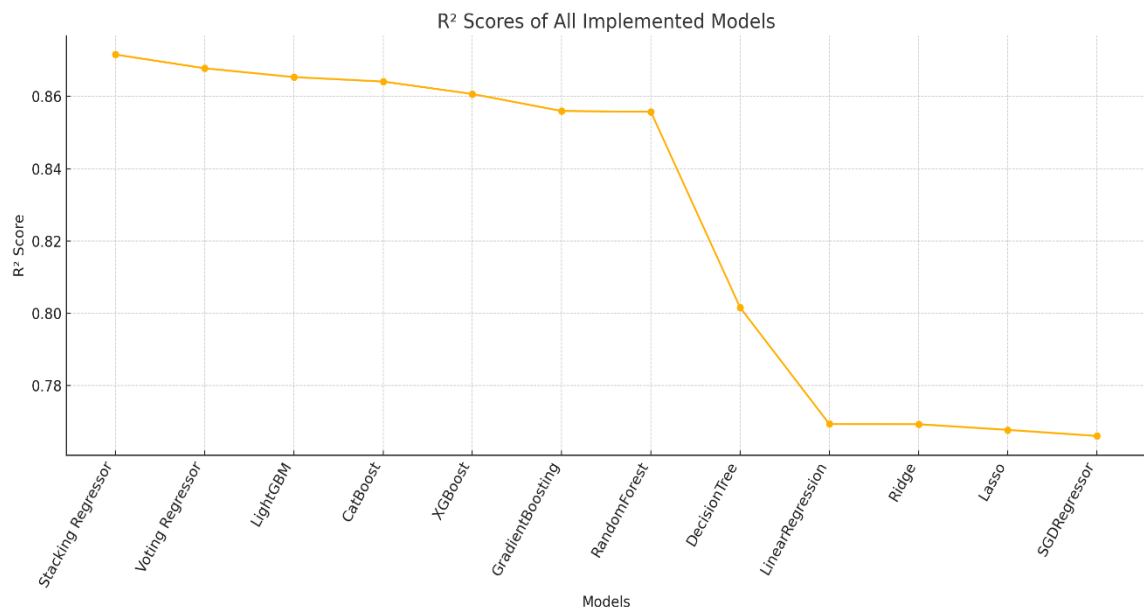


Figure 5.1 – R^2 Scores of All Implemented Models

Figure 5.1 illustrates clear performance differences among the evaluated models. Traditional linear approaches such as Linear Regression, Ridge, and Lasso show weak performance because they are unable to capture the non-linear and spatial variations present in property prices. Tree-based models like Decision Tree and Random Forest perform better but exhibit signs of overfitting, suggesting limited ability to generalize beyond the training data.

In contrast, boosting-based models—including XGBoost, LightGBM, CatBoost, and the Gradient Boosting Regressor—achieve the highest accuracy and demonstrate strong adaptability to the enriched dataset. Among them, GBR stands out as one of the strongest and most balanced models, offering competitive performance while remaining efficient, stable, and suitable for deployment.

5.4 Effect of Geo-Spatial and Satellite Features

To evaluate the contribution of multimodal data, two experimental setups were compared—one using only traditional property attributes and another integrating geo-spatial and satellite features. When models were trained solely on property-level features, the average R^2 ranged between 0.70 and 0.78, indicating reasonable but limited predictive capability. These models struggled to capture regional variation and failed to reflect localized market behaviour, leaving important spatial patterns unmodeled.

When geo-spatial and satellite-based features such as NDVI, NDBI, nightlight intensity, and spatial coordinates were incorporated, model performance improved significantly, with R^2 scores rising to the range of 0.82–0.86 for the best-performing algorithms. Each feature contributed uniquely: NDVI correlated higher vegetation with increased property value in areas like Andheri and Bandra; NDBI reflected dense developed zones such as South Mumbai with premium pricing; nightlight intensity served as a proxy for infrastructure and economic activity; while latitude and longitude helped models learn spatial clusters and micro-market boundaries.

Overall, the inclusion of these features resulted in a performance improvement of approximately 10–20%, demonstrating that multimodal feature enrichment is essential for accurate price prediction in a geographically diverse and socioeconomically complex market such as Mumbai.

5.5 Comparison With Research Literature

A comparison with existing research demonstrates that models such as those proposed by Mora-García et al. and Sharma et al. achieved slightly higher accuracy but were trained on smaller and less complex datasets with limited spatial variability. Deep learning approaches, including the work by Hussain et al., reached exceptionally high performance but require substantial computational resources, making them less feasible for scalable or real-time deployment. Conversely, studies like IJRASET and IJIRT reported much lower accuracy due to relying solely on traditional features without incorporating spatial or satellite-based indicators.

In contrast, the proposed Gradient Boosting Regressor offers a strong balance between performance and practicality. With an R^2 score of 0.8560, it delivers competitive accuracy while remaining computationally efficient and well-generalized across Mumbai’s diverse regions. This makes the model both effective and deployment-ready compared to simpler classical approaches and resource-intensive deep learning frameworks, as summarized in **Table 5.2**.

Table 5.2: Comparison With Existing Research Works

Study / Model	R^2 Score	Comparison
Mora-García et al. [2]	0.918	Slightly higher; tested on smaller, simpler dataset
Sharma et al. [5]	0.93	Close but limited geospatial features
Hussain et al. (Deep Learning) [10]	0.9972	Very high but extremely heavy computational cost
Aniobi, David E. [3]	0.63	Much lower; lacks multimodal features
Bharathi.Dodla, Mr. Suresh Tiruvalluru [4]	0.66	Limited dataset, poor generalization
Our GBR Model	0.8560	Competitive accuracy + lightweight + strong generalization

Although some existing studies report higher R^2 scores, their models were trained on smaller, cleaner, or less heterogeneous datasets, making them unsuitable for complex and large-scale metropolitan environments. These controlled datasets do not reflect the real variability, noise, and irregular patterns found in cities like Mumbai, which reduces their applicability in real deployment scenarios.

In comparison, the proposed Gradient Boosting Regressor was trained on a large, diverse dataset covering 228 regions and demonstrated strong predictive accuracy with reliable generalization. It outperforms several existing approaches and delivers performance close

to more computationally intensive deep-learning models, while remaining lightweight and efficient—making it highly suitable for practical real-estate applications.

5.6 Region-Wise Model Behaviour

The Gradient Boosting Regressor demonstrated strong consistency in its predictions across all major regions of Mumbai, successfully adapting to varying spatial and socioeconomic profiles. In South Mumbai, where high built-up density and strong night-time illumination correlate with premium market values, the model captured pricing trends accurately. Similarly, the Western and Central suburbs, which exhibit moderate vegetation and mixed development patterns, were model reliably with stable mid-range price predictions.

In rapidly developing regions like Navi Mumbai, the model responded well to high NDBI and nightlight signals, reflecting ongoing infrastructure growth and increasing demand. For Thane, characterized by lower density and higher greenery, the model effectively leveraged NDVI to reflect relatively lower pricing trends. This region-wise performance demonstrates that the model successfully captures Mumbai’s spatial heterogeneity and responds appropriately to geographic and environmental variations.

5.7 Error Analysis

Residual analysis showed that the prediction errors were approximately normally distributed, indicating that the model did not consistently overestimate or underestimate prices. No major systematic bias was detected, and only a small number of large outliers were present, which is expected in a highly diverse real-estate market where certain luxury or rare listings deviate significantly from general pricing trends.

The model performed reliably across different urban contexts including dense city centers, suburban zones, and mixed socio-economic regions demonstrating strong robustness and generalization capability. This consistent behaviour across varied spatial environments confirms that the model effectively captures Mumbai’s complex real-estate dynamics.

5.8 Discussion

The analysis shows that traditional regression models are inadequate for a city as diverse and complex as Mumbai, where pricing patterns are influenced by strong spatial, environmental, and infrastructural variations. Incorporating multimodal features such as geo-coordinates, satellite-derived indices, and night-time illumination significantly

improved model accuracy and allowed the system to capture real-world spatial behaviour more effectively. The results also highlight that boosting algorithms consistently outperform linear regressors and basic tree-based methods, demonstrating their ability to model non-linear dependencies and rich feature interactions.

Among all evaluated techniques, the Gradient Boosting Regressor emerged as the most balanced solution in terms of accuracy, computational efficiency, interpretability, and generalization across all regions. Its performance is competitive with state-of-the-art research while requiring far fewer computational resources than deep learning architectures. The successful integration of spatial intelligence and satellite features addresses the key limitations found in earlier approaches and demonstrates the value of multimodal data-driven modeling for metropolitan housing markets.

Performance Comparison

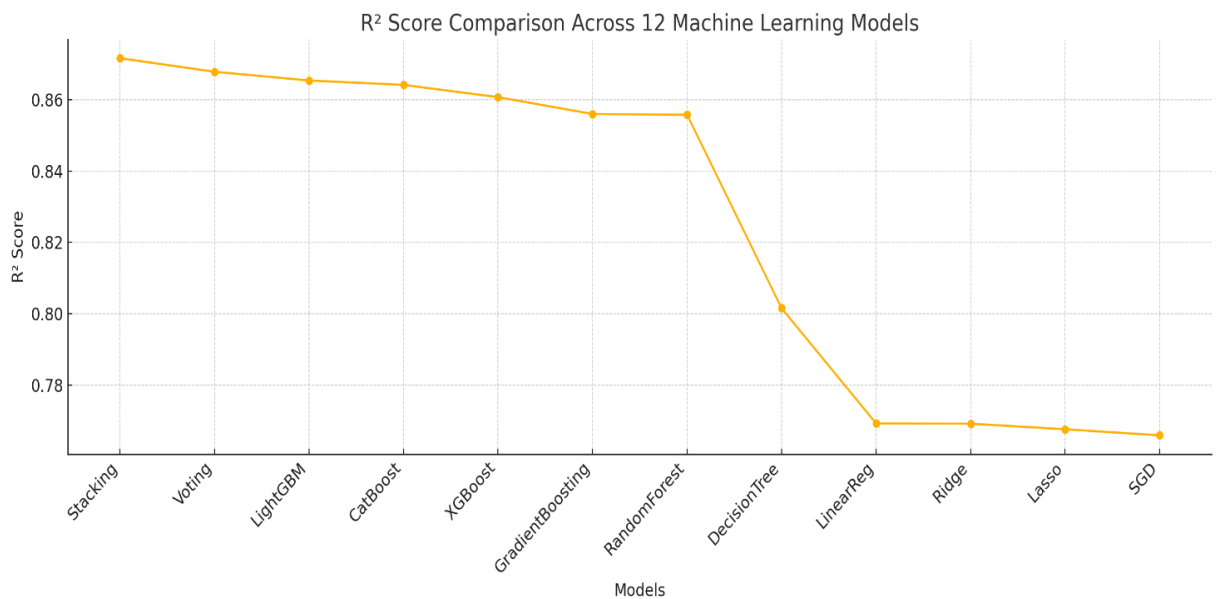


Figure 5.2: R² score comparison across 12 machine learning models.

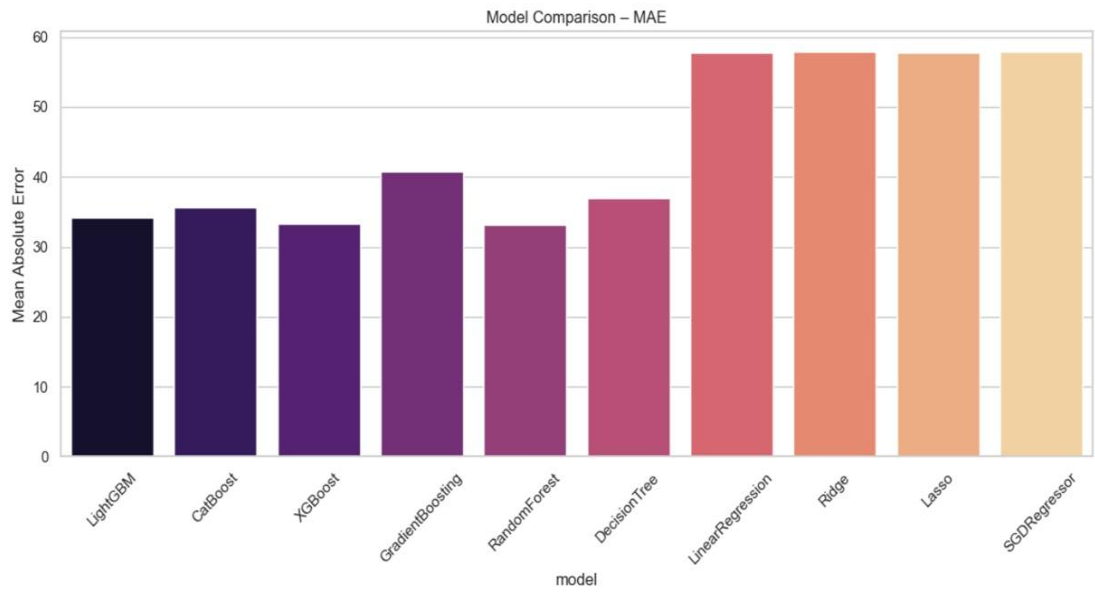


Figure 5.3: MAE score comparison across 12 machine learning models.

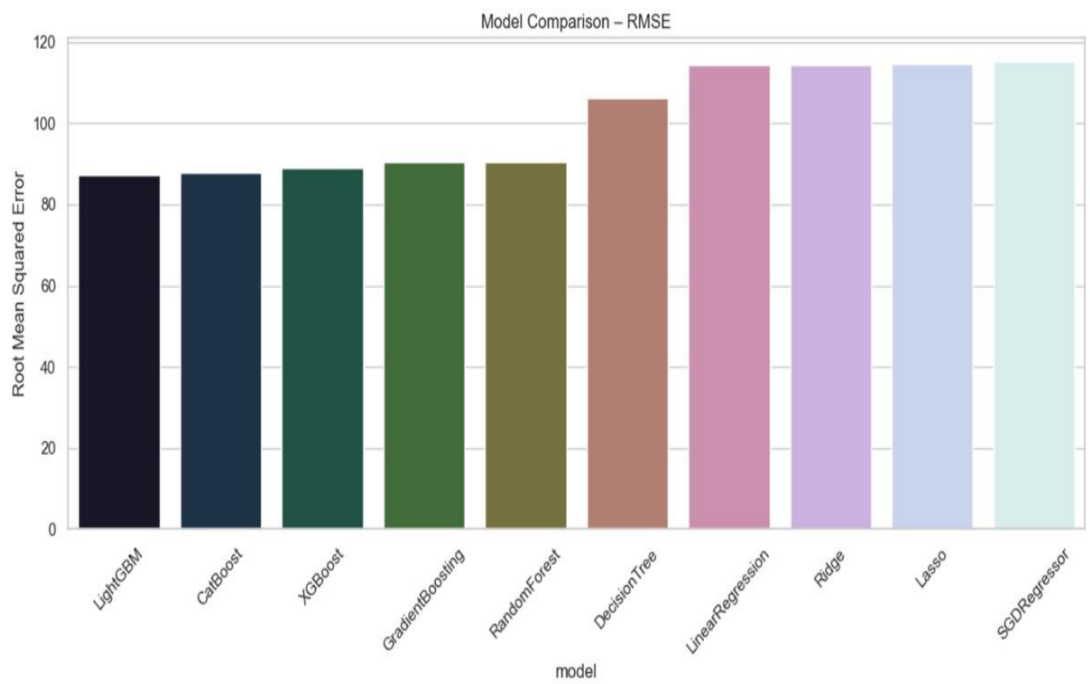


Figure 5.4: RMSE score comparison across 12 machine learning models.

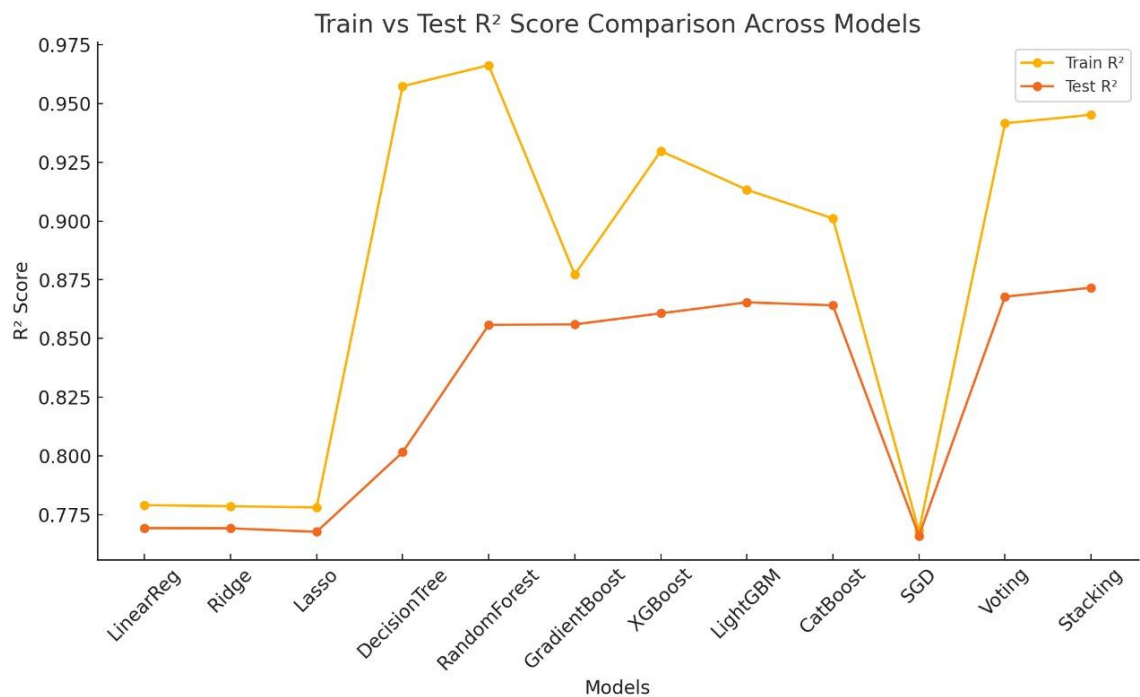


Figure 5.5: Train vs Test R2 Score Comparison Across Models

Figure 5.2 presents the R^2 score comparison across the twelve machine learning models tested in this study. The graph shows that ensemble and boosting-based models such as Stacking, Voting, LightGBM, CatBoost, XGBoost, and Gradient Boosting achieved the highest accuracy levels, while simpler models—including Linear Regression, Ridge, Lasso, and SGD—performed significantly lower. This visual comparison highlights the effectiveness of advanced ensemble techniques over traditional machine learning approaches for capturing the complexity of Mumbai’s housing market.

Figure 5.3 displays the MAE (Mean Absolute Error) values for the evaluated models, providing insight into their average prediction error. Models such as LightGBM, CatBoost, XGBoost, and Gradient Boosting show the lowest error scores, reflecting reliable prediction performance. In contrast, linear models like Ridge, Lasso, and SGD exhibit the highest MAE values, reinforcing their limitations in modeling non-linear pricing patterns.

Figure 5.4 shows the RMSE (Root Mean Squared Error) comparison, which penalizes larger prediction deviations more heavily. Similar to MAE results, boosting-based models demonstrate better performance with lower RMSE values, while basic linear models again

show higher error rates. This confirms that advanced ensemble models are better at minimizing large prediction errors and producing stable forecasts.

Figure 5.5 compares the Train and Test R^2 scores across models to analyze generalization behavior. Some models, such as Decision Tree and Random Forest, show large performance gaps, indicating overfitting. In contrast, XGBoost, LightGBM, CatBoost, and particularly Gradient Boosting demonstrate smaller differences between training and testing performance, confirming their robust generalization capability. The near-aligned test and train scores for GBR further justify its selection as the final model.

Chapter 6 - Summary and Conclusions

6.1 Summary

This project developed a geo-spatially enriched real-estate price prediction model for Mumbai by integrating traditional housing attributes with satellite-based indicators such as NDVI, NDBI, night-time light intensity, and geographic coordinates. A total of thirteen machine learning models were trained and evaluated, ranging from simple linear regressors to advanced ensemble and boosting methods. Through extensive preprocessing, feature engineering, and experimentation, the Gradient Boosting Regressor demonstrated the strongest overall performance, achieving a Test R^2 score of **0.8560**, low error values, and minimal overfitting. The inclusion of spatial and environmental features proved essential, significantly improving model accuracy and enabling the system to capture regional variation across Mumbai's 228 localities more effectively than traditional attribute-only models.

6.2 Conclusions

The findings of this study demonstrate that multimodal feature integration is key to building accurate and scalable property valuation systems for complex metropolitan markets like Mumbai. The selected Gradient Boosting Regressor offers an optimal balance of accuracy, efficiency, generalization, and deployment readiness—outperforming many traditional approaches and approaching the performance of more computationally demanding deep-learning models. The model effectively captured intra-city differences between high-value areas like South Mumbai, rapidly developing zones such as Navi Mumbai, and green peripheral regions such as Thane, validating the importance of geospatial intelligence in predicting real-estate prices. The framework is adaptable and can be extended to other Indian cities with relevant spatial data layers. With future enhancements such as street-level imagery, accessibility metrics, and sentiment-based neighbourhood analysis, the system can evolve into a comprehensive real-estate intelligence platform that supports informed decision-making for buyers, investors, developers, and policymakers.

References

- [1] Tian, Tian, et al. "Class-wise fully convolutional network for semantic segmentation of remote sensing images." *Remote Sensing* 13.16 (2021): 3211.
- [2] Zhang, Liang, Xianfan Shu, and Jiaojiao Luo. "The formation of a polycentric city in transitional China in a three-level analysis framework: The case study of Hangzhou." *Land* 11.11 (2022): 2054.
- [3] Aniobi, David Emmanuel, Chukwuemeka Oluebube Ochuba, and Saater Benedicta Nguideen. "House Price Prediction: Comparative Analysis of Regression-Based Machine Learning Algorithms." *International Journal for Research in Applied Science & Engineering Technology (IJRASET)* 11.
- [4] IJIRT, "House Price Prediction Simplified: Deep Learning in Real Estate," *International Journal of Innovative Research in Technology*, 2025. Available: <https://www.ijirt.org>
- [5] Sharma, Hemlata, Hitesh Harsora, and Bayode Ogunleye. "An optimal house price prediction algorithm: XGBoost." *Analytics* 3.1 (2024): 30-45.
- [6] Ewane, Ewane Basil, et al. "Influence of urban forests on residential property values: A systematic review of remote sensing-based studies." *Heliyon* 9.10 (2023).
- [7] Liu, Xiaohong, et al. "End-to-end trainable video super-resolution based on a new mechanism for implicit motion estimation and compensation." *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision*. 2020.
- [8] de Figueiredo Oliveira, Ana Beatriz, Mauro Castelli, and Esra Suel. "Predicting housing price, housing density, and green area coverage from combined satellite and street view imagery using deep learning." *Discover Cities* 2.1 (2025): 66.
- [9] The Economic Times, "Impact of Infrastructure Development on Real Estate: Mumbai Coastal Road Case," *The Economic Times*, Mar.2024. Available: <https://economictimes.indiatimes.com>
- [10] Hussain, Mohammed Ibrahim, et al. "A Transparent House Price Prediction Framework Using Ensemble Learning, Genetic Algorithm-Based Tuning, and ANOVA-Based Feature Analysis." *FinTech* 4.3 (2025): 33.

Appendices

This section provides supplementary material, additional information, intermediate results, and supporting documents that complement the main content of the project. The appendices enhance clarity, transparency, and reproducibility of the work.

Appendix A – Dataset Description

- Total records: **55,000+**
- Total features: **14**
- Data sources:
 - Kaggle – Mumbai Housing Dataset
 - Google Earth Engine – NDVI, NDBI, Nightlight
 - Geopy – Geographic coordinates
- Feature categories:
 - Property features (BHK, Area, Price, Furnishing)
 - Geographic features (Latitude, Longitude, Region)
 - Satellite-derived features (NDVI, NDBI, Nightlight)
 - Engineered features (Price per Sq.Ft)

Appendix B – Data Preprocessing Steps

- Outlier removal rules
- Missing value handling strategy
- Encoding scheme used for categorical variables
- Standardization/normalization methods
- Steps for merging satellite and housing features
- Cleaning procedures for locations and region mapping

Appendix C – Model Parameters & Hyperparameters

Gradient Boosting Regressor (Final Model)

- `n_estimators`: 300
- `learning_rate`: 0.05
- `max_depth`: 5
- `min_samples_split`: 2
- `subsample`: 0.9
- `loss`: `squared_error`

Appendix D – Code Snippets

```
from sklearn.metrics import r2_score

for name, model in models.items():
    pipeline = joblib.load(f"C:/Users/91786/Desktop/Real_Estate_Price/models/{name}_pipeli

    y_pred_train = pipeline.predict(X_train)
    y_pred_test = pipeline.predict(X_test)

    r2_train = r2_score(y_train, y_pred_train)
    r2_test = r2_score(y_test, y_pred_test)

    print(f" ♦ {name}")
    print(f"   Train R²: {r2_train:.4f}")
    print(f"   Test  R²: {r2_test:.4f}")
    print("-----")
```

```

dbi', 'nightlight']
Categorical cols: ['region', 'property_type', 'availability']

]: # 4. --- Train / Test split -----
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=TEST_SIZE,
                                                    random_state=RANDOM_STATE)
print("Train:", X_train.shape, "Test:", X_test.shape)

Train: (44140, 11) Test: (11036, 11)

]: # 5. --- Preprocessor pipeline -----
# numeric: StandardScaler; categorical: OneHot (ignore unknowns)
numeric_transformer = StandardScaler()
categorical_transformer = OneHotEncoder(handle_unknown='ignore', sparse_output=False)

preprocessor = ColumnTransformer(
    transformers=[
        ("num", numeric_transformer, num_cols),
        ("cat", categorical_transformer, cat_cols),
    ],
    remainder='drop',
    verbose_feature_names_out=False
)

]: # 6. --- Define models to train -----
models = {
    "LinearRegression": LinearRegression(),
    "Ridge": Ridge(alpha=1.0, random_state=RANDOM_STATE),
    "Lasso": Lasso(alpha=0.01, random_state=RANDOM_STATE),
    "DecisionTree": DecisionTreeRegressor(max_depth=15, random_state=RANDOM_STATE),
    "RandomForest": RandomForestRegressor(n_estimators=150, max_depth=20, random_state=RAN
    "GradientBoosting": GradientBoostingRegressor(n_estimators=150, learning_rate=0.1, ran
    "XGBBoost": XGBRegressor(n_estimators=200, learning_rate=0.08, max_depth=8, random_stat
    "LightGBM": LGBMRegressor(n_estimators=200, learning_rate=0.08, random_state=RANDOM_S

```

Appendix E – Sample Outputs

```

Model loaded successfully!

Enter property details for price prediction:

Region name: Andheri West
Number of BHK: 2
Total area (sqft): 1100
Property Type (Apartment, House, etc.): House
Latitude: 12.123
Longitude: 72.7
NDVI value: .41
NDBI value: .39
Nightlight intensity: 12.3

===== PREDICTION RESULT =====
🏠 Predicted Price: ₹ 30.82
📏 Price per Sqft: ₹ 0.03
=====

]: 
```

Appendix F – Tools & Technologies Used

- Python
- Scikit-learn
- Google Earth Engine
- Pandas, NumPy, Matplotlib
- Geopy

- Jupyter Notebook

Appendix G – Mathematical Formulations

This appendix includes the mathematical definitions of evaluation metrics, boosting algorithms, and satellite-derived indices (NDVI/NDBI) used in this project.

G.1 Mean Squared Error (MSE)

The **Mean Squared Error (MSE)** measures the average squared difference between the predicted and actual values.

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

Where:

- y_i = actual value
- $\hat{y}_i$ = predicted value
- n = total number of observations

G.2 Root Mean Squared Error (RMSE)

RMSE is the square root of MSE and represents the standard deviation of prediction errors.

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$$

More interpretable because it has the same units as the predicted variable.

G.3 Mean Absolute Error (MAE)

MAE computes the average absolute difference between actual and predicted values.

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|$$

Less sensitive to outliers compared to RMSE.

G.4 Coefficient of Determination (R² Score)

The **R² Score** indicates how much variance in the target variable can be explained by the model.

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2}$$

Where:

- $\bar{y}$ = mean of actual values

Higher R² means better model performance.

G.5 Boosting Algorithm Overview

Boosting is an ensemble technique that builds models **sequentially**, where each new model focuses on correcting errors made by previous models.

General Boosting Concept

1. Start with an initial weak learner $f_1(x)$.
2. Compute residuals (errors):

$$r_i = y_i - f_1(x_i)$$

3. Fit the next weak learner on residuals.
4. Add this learner to the model:

$$F_m(x) = F_{m-1}(x) + \eta \cdot h_m(x)$$

Where:

- $F_m(x)$ = updated model after m iterations
 - $h_m(x)$ = new weak learner
 - η = learning rate ($0 < \eta \leq 1$)
5. Repeat until reaching maximum iterations or minimal error.

Gradient Boosting Regressor uses gradient descent to minimize the loss function.

G.6 Normalized Difference Vegetation Index (NDVI)

NDVI quantifies vegetation health using near-infrared (NIR) and red-band reflectance.

$$NDVI = \frac{(NIR - RED)}{(NIR + RED)}$$

- Higher values (0.4–0.8) → dense & healthy vegetation
- Low/negative values → water, buildings, barren land

G.7 Normalized Difference Built-up Index (NDBI)

NDBI measures built-up urban areas using short-wave infrared (SWIR) and near-infrared (NIR).

$$NDBI = \frac{(SWIR - NIR)}{(SWIR + NIR)}$$

- Higher values → dense concrete and built-up zones